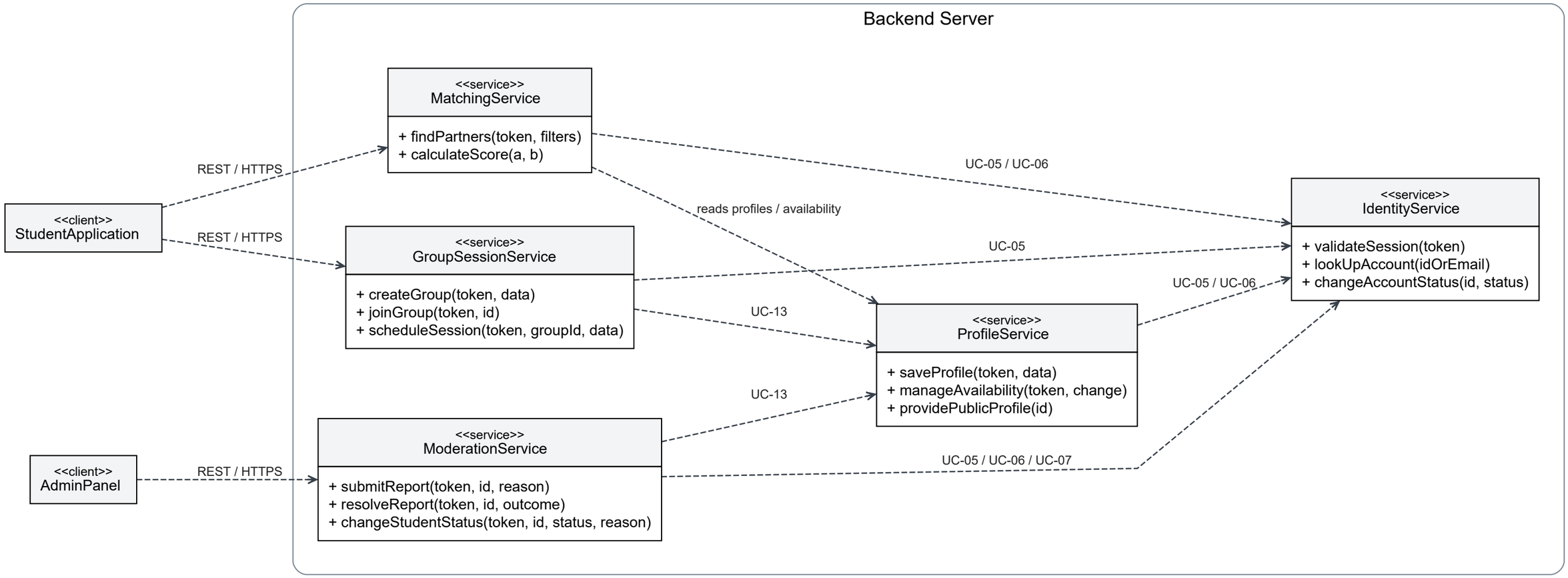
System services and clients

Three communicating applications; four backend modules



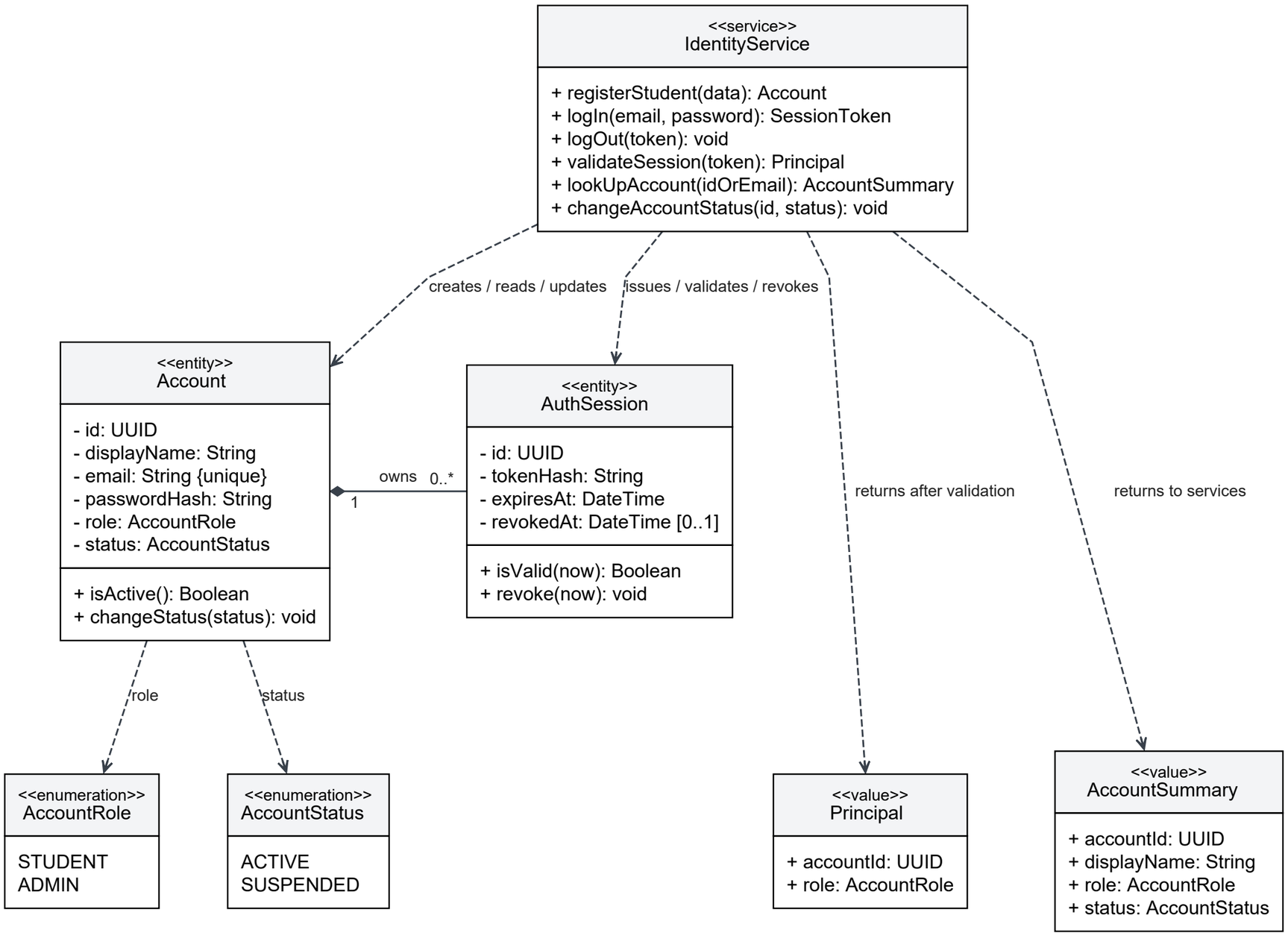
Student Application and Admin Panel call the Backend Server over REST/HTTPS. Internal service calls implement Julia's UC-05, UC-06, UC-07 and UC-13.

Selected operations and main client flows are shown. Both clients also use IdentityService; the student client also uses ProfileService and ModerationService. ProfileService and MatchingService share one module.

All protected operations validate the caller and check permissions. Internal account lookup and status changes are trusted service operations. Full operations appear in the following views.

Identity & Access

Accounts, roles and authentication sessions



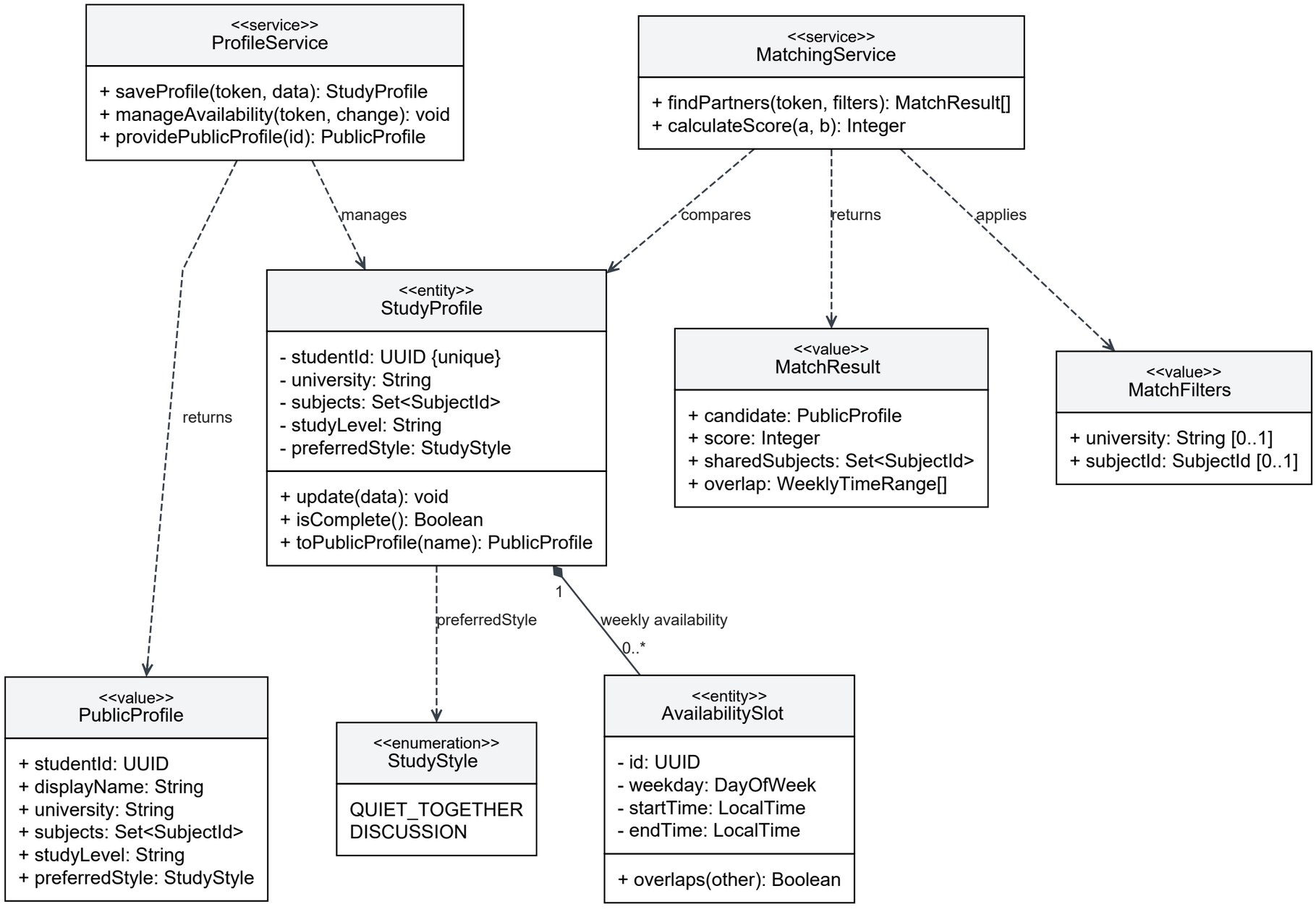
Each account has one role and one status. Public registration creates students only. A student may exist before completing a study profile.

AuthSession is a login session, not a study session. Validation checks token expiry/revocation and current account status; logout revokes the current session. Suspension revokes every login session.

AccountSummary supplies identity, display name, role and status to services. Credentials and email are excluded. Solid lines show associations, filled diamonds composition, and dashed arrows dependencies.

Profile & Matching

Study preferences, recurring availability and calculated results



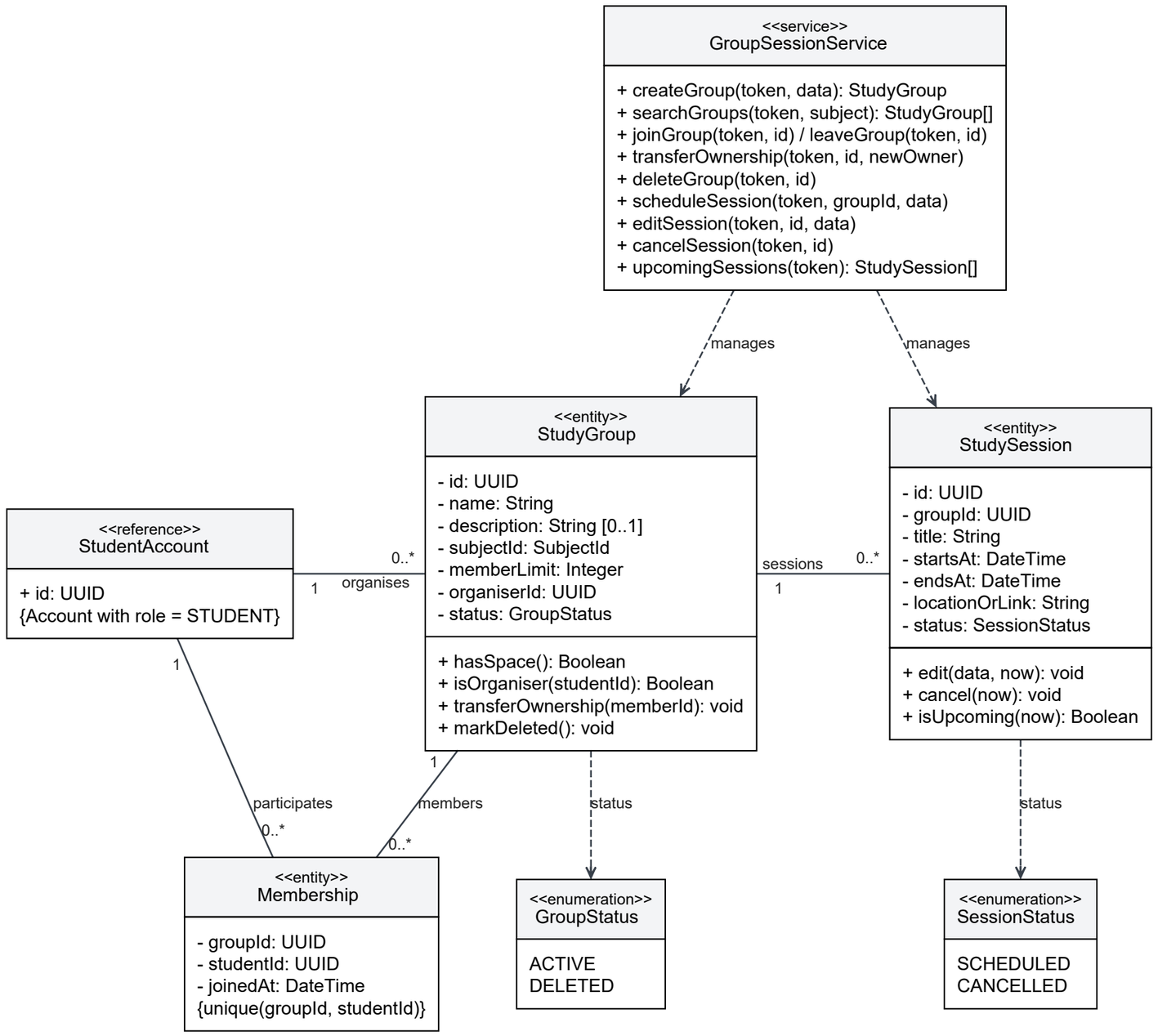
A student has at most one StudyProfile. A saved profile needs a university, subjects, programme study level and preferred style. Slots recur weekly in Europe/Warsaw and overlapping slots are merged.

Matching requires an active account, a shared subject and positive availability overlap. Score = distinct shared subjects + same university + same study level + same style, with one point for each equality.

Selected filters use AND; equal scores use student ID. MatchResult is temporary and creates no membership. PublicProfile excludes email and credentials; WeeklyTimeRange stores weekday and start/end times.

Group & Session

Group membership, ownership and scheduled study sessions



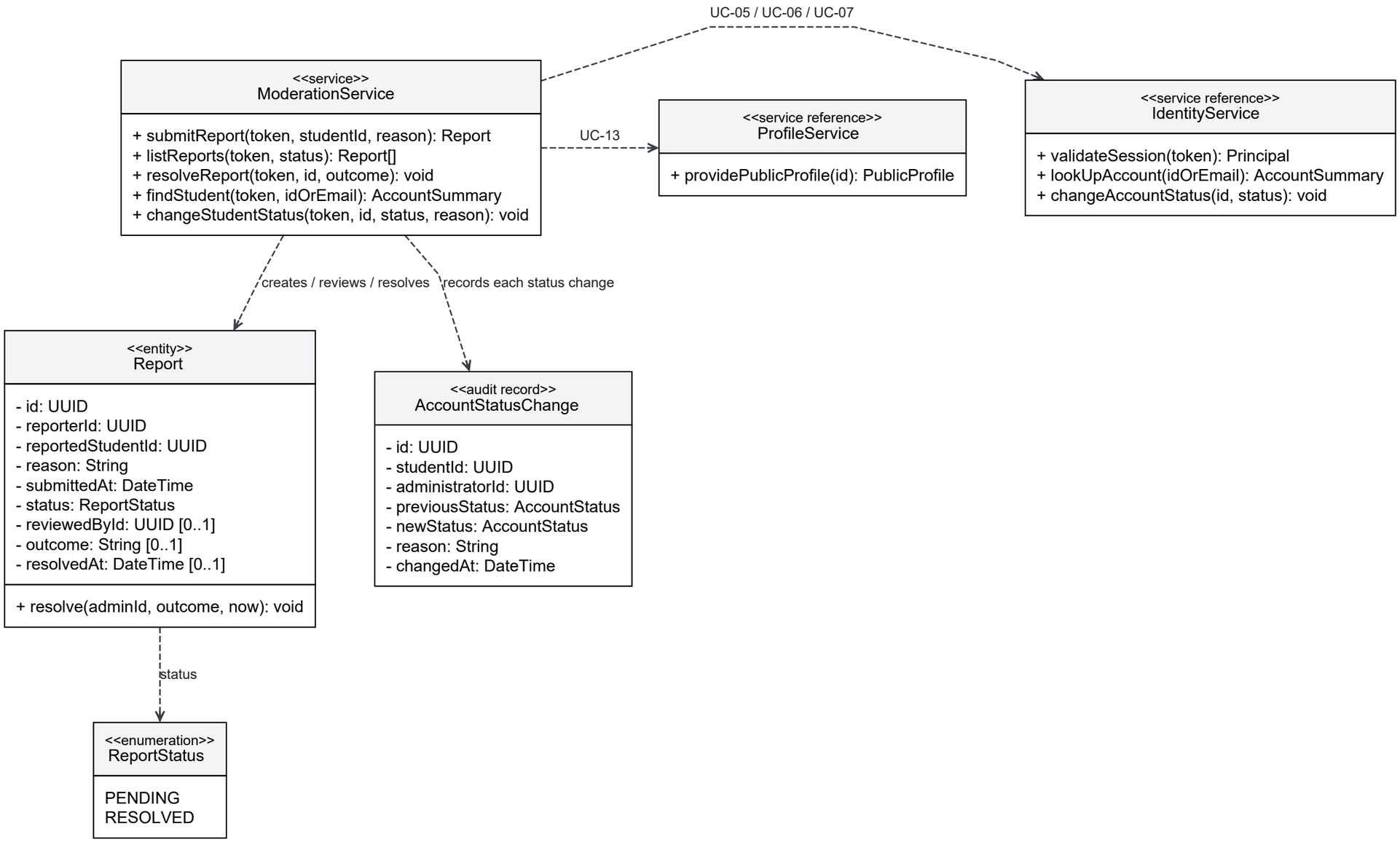
An active group has exactly one organiser who is also a member. StudentAccount is a reference to an Account with role STUDENT, not a separate entity. Organiser is a role within a particular group.

Membership is unique per student/group; capacity includes the organiser and is checked atomically. An organiser must transfer ownership to an active member or delete the group before leaving.

Deletion removes memberships, hides the group and cancels future sessions. Only organisers schedule/edit/cancel; only current members view details. Cancelled future sessions remain labelled unless the group was deleted.

Moderation

Reports and account-status audit records



The reporter and reported student must differ. A pending report has no reviewer, outcome or resolution time. Resolution records all three and changes status to RESOLVED; only administrators can review reports. Resolving a report does not suspend an account. Status changes can occur without a report. Each AccountStatusChange records the acting administrator, target student, old/new status, reason and time. ModerationService validates permissions, changes status through IdentityService and records the audit entry atomically. Suspension preserves profiles and memberships but blocks login and protected actions.